The Link Function Problem in Psychological Interaction Testing

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Abstract

Equal differences on one scale become different differences on another scale. This is crucial for interaction testing because an interaction is a difference between differences, and in psychological research the observed response and the construct of interest are often expressed on different scales. Therefore, the same data can show an interaction on one scale and not on another. Generalized Linear Models make this problem explicit, as the link function defines the scale on which the model without an interaction term is additive. Choosing a link thus defines what ‘no interaction’ means, while the family describes the nature of the response variable. The paper has two connected parts. First, a preregistered review of recent articles from five leading psychology journals shows that interaction testing is routinely performed in research, that the link function is most often unspecified, and that constrained outcomes are routinely analyzed with identity-link models without justifying the assumption of equal effects on the observed scale. Second, worked examples and simulations develop three common response formats: forced-choice accuracy with a chance floor, within-family link choices (logit vs probit), and sum scores as a measurement case, and quantify how a plausible-looking link can turn an additive structure on one scale into a statistically significant interaction on another. A final diagnostic simulation shows that residual checks, a link test, and AIC comparisons give uneven protection and do not by themselves decide which scale the scientific claim concerns. Interaction claims remain underspecified until researchers state, justify, and probe the scale on which no interaction is assumed.

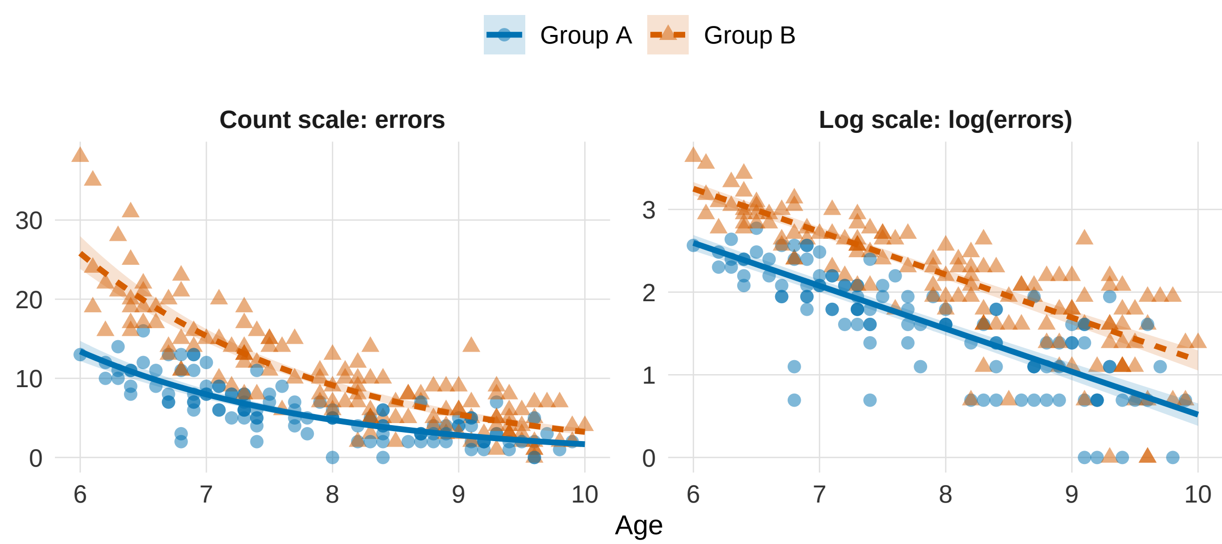
Keywords: interactions, moderation, generalized linear models, link functions, model misspecification

The Link Function Problem in Psychological Interaction Testing

We start with a very simplified example. Two populations of children (e.g., clinical vs neurotypical) differ in their overall error rates on a task, and errors decrease with age in both. Figure 1 samples simulated data from such a scenario. In the left panel (observed response), errors decline with age in both groups, and the difference between groups is much larger at age six than at age ten. Is there an age-by-group interaction here?

Figure 1

Simulated error counts for two groups of children aged 6 to 10, shown on the observed count scale (left) and on the log scale (right). The two groups have exactly the same age effect on the log of the expected error count, so on the log scale the two fitted lines are parallel, with a constant group gap at every age. Mapped back onto the count scale, the same model produces converging curves, so the group difference shrinks with age. Whether this pattern is an interaction depends on the scale on which effects are assumed to add: on the count scale the group difference changes with age; on the log scale it does not.



Because an interaction is a difference between differences, the question is whether the group difference itself changes with age. On the observed count scale, the group effect does appear to shrink as age increases, so it looks like an interaction. But looking at the same data on the log-scale (Figure 1, right panel), both groups have the same age slope, meaning their fitted lines are parallel. A constant difference on the log scale corresponds to a constant ratio on the count scale, so the absolute gap between groups is proportional to the baseline and shrinks as errors decline with age. Whether we call this an interaction is not a fact about the data, it depends on which scale we assume effects add up linearly on. Additive on the log scale means multiplicative (and non-parallel) on the count scale, and vice versa.

This ambiguity extends far beyond this single illustration, as interaction hypotheses are widespread in psychological research and often represent the central target of inquiry (Domingue et al., 2024; McCabe et al., 2022; Rohrer & Arslan, 2021). Whenever researchers ask whether an effect is moderated, or whether it holds only under certain boundary conditions, they are asserting that the impact of one variable depends on another. This count example therefore demonstrates that the reality of such a claim can shift entirely based on the scale on which those differences are computed, as equal differences on one scale are different differences on another scale.

In psychology, this scale dependence has a characteristic source in measurement. Researchers routinely conceptualize constructs such as cognitive ability or symptom severity as continuous latent dimensions (Embretson & Reise, 2000; Wagenmakers et al., 2012). Those dimensions are rarely observed directly; what reaches the data are bounded, discrete quantities, such as counts, ratings, and accuracy scores that carry floors, ceilings, and chance levels. Each measurement scale imposes its own mapping from the latent dimension to the observed response. Where the mapping compresses, a constant effect on the latent dimension appears as a shrinking effect on the observed response: a group or condition closer to a bound has less room to move, and can look less affected by a manipulation, than a group or condition farther away from bounds, even if the underlying data-generating process remains the same at all levels of the scale.

Generalized linear models (GLMs) make this mapping explicit and give it a standard vocabulary (McCullagh & Nelder, 1989; Nelder & Wedderburn, 1972), separating two choices that are easy to conflate: the outcome family, which describes the response, and the link function, which sets the scale on which effects combine (Box 1).

Box 1: Generalized linear models: outcome family and link function

The *outcome family* is the distribution assumed for the response. It defines which values the outcome can take and how observations vary around their expectation. The *link function*, $g(\cdot)$, maps the expected outcome, $\mu_i = E(Y_i)$, onto the linear predictor, η_i, the scale on which the model is additive. With two predictors and their interaction term,

$$g(\mu_i) = \eta_i, \qquad \eta_i = \beta_0 + \beta_1 X_i + \beta_2 Z_i + \beta_3 X_i Z_i.$$

The inverse link, $g^{-1}(\cdot)$, maps this additive structure back onto the expected response. Its familiar role is to keep expected values within the admissible range of a constrained outcome: $\exp(\eta)$ keeps expected counts positive, and the inverse logit keeps probabilities between 0 and 1. Less obviously, the link thereby defines the scale on which effects add, and therefore what the absence of an interaction, $\beta_3 = 0$, means in the fitted model. The product-term coefficient β_3 is therefore an interaction on the link scale. A response-scale interaction is a different quantity: a difference between expected differences on the observed outcome, for example, whether the group difference in expected errors changes with age in Figure 1. Unless the link is the identity, the two quantities can disagree (Ai & Norton, 2003; McCabe et al., 2022; Mize, 2019). Each family has a canonical link, but other links are legitimate. A Gaussian linear model is the special case that combines a Gaussian family with an identity link, so the additive scale and the response scale coincide and the scale question is invisible. The model behind Figure 1 is a Poisson family with a log link. The family respects that errors are non-negative counts, and the link places additivity on the log scale.

Often the outcome family is chosen with care while the link function is left at its software default, fixing the scale of additivity without deliberation. Two models can share a family and still place additivity on different scales. For example, a Gamma family suits reaction times data; within that family, a log link makes effects additive in log milliseconds, a proportional change, and the canonical inverse link makes them additive in reciprocal means, a change in speed. A binomial family fits correct-versus-incorrect responses and leaves the link open among logit, probit, chance-corrected, and other forms.

The scale dependence of interactions has long been recognized. Loftus showed that interactions involving response probabilities are meaningful only relative to a mapping between observed and theoretical scales (Loftus, 1978). Cox formalized interaction as a model-relative concept tied to transformation and interpretation (Cox, 1984). Wagenmakers and colleagues distinguished removable from nonremovable interactions, showing that some depend entirely on the measurement scale (Wagenmakers et al., 2012). Busemeyer showed that conclusions about combination rules depend on the measurement model (Busemeyer & Jones, 1983), and Lubinski and Humphreys illustrated how model specification choices can mimic moderator effects (Lubinski & Humphreys, 1990). More recently, Rohrer and Arslan argued that interaction tests often provide precise answers to vague questions when the estimand and scale are left unspecified (Rohrer & Arslan, 2021), and Rimpler and colleagues showed that product terms in linear models can mismatch the functional form implied by a substantive conditional-effect claim (Rimpler et al., 2026). In GLMs specifically, product-term coefficients are not generally equal to response-scale interactions, so interaction effects should be evaluated as marginal effects or response-scale contrasts (Ai & Norton, 2003; McCabe et al., 2022; Mize, 2019; Rohrer & Arel-Bundock, 2026). Domingue and colleagues demonstrated that outcome-model misspecification can produce false interaction findings for binary, count, and censored outcomes (Domingue et al., 2024), and Liddell and Kruschke showed that treating ordinal responses as metric can inflate error rates for interactions (Liddell & Kruschke, 2018). Czado and Santner, and later Yu and Huang, studied the inferential consequences of link misspecification in binary GLMs and GLMMs (Czado & Santner, 1992; Yu & Huang, 2019). Interaction estimates are also fragile for ordinary reasons, including small effects, low reliability, and limited power (Baranger et al., 2023; Castillo et al., 2026).

These contributions identify adjacent parts of a shared problem. None of them, however, directly addresses link-choice-driven interaction artifacts in GLMs for bounded or chance-limited psychological outcomes, the case in which the outcome family is broadly plausible but the link still sets a disputable no-interaction baseline. An interaction on a scale other than the generating one is not a false positive in a strict statistical sense; it is a genuine mathematical property of that scale (Cox, 1984; Domingue et al., 2024; Loftus, 1978; McCabe et al., 2022). We define a pseudo-interaction as an interaction contrast that is zero on the known generating scale but nonzero on another fitted scale because the two scales encode different additivity baselines. This label is available in simulations because the generating process is known. In empirical data, where the generating scale is unknown, we instead describe interaction conclusions as link-sensitive, conditional on the fitted scale, or unstable across plausible links. Some interactions survive any monotonic rescaling, crossover interactions being the classical example (Wagenmakers et al., 2012). Many others do not, and the present paper focuses on these: the incremental contribution is to separate outcome family, link function, and target metric in a single framework for interaction testing, combine this separation with a preregistered review documenting the reporting gap, and quantify the inferential risk through simulations of common psychological response formats.

The present article addresses these issues in two parts. The first is a preregistered descriptive review of recent empirical articles in five prominent psychology journals, quantifying how often interactions are tested, how often the outcome family and link function are made explicit, and which response formats predominate. The second develops the link function problem in three common response formats: forced-choice accuracy with a chance floor, sum scores, and within-family link choices such as logit versus probit. Using simulations, it quantifies how often an inadequate link turns a purely additive structure into a statistically significant interaction, and whether standard diagnostics detect the misspecification. Table 4 summarizes these insights into a practical workflow for applied research.

Technical details are reported in Supplement A; a broader precomputed sensitivity atlas varying sample size, main-effect magnitude, scale location, and selected design parameters is reported in Supplement B.

Current Practice in Psychological Interaction Testing: A Preregistered Review

The review asks whether the scale-dependence problem is visible in current reporting practice: how often psychological articles test interactions, and how often they make the outcome family and link function explicit enough for a reader to know which interaction claim was tested. To our knowledge, it is the first review to quantify link-function reporting in psychological interaction tests. The coding is deliberately descriptive, based only on what a published report makes observable, so it documents reporting practice rather than the correctness of any substantive conclusion.

Review Method

The protocol was preregistered before any record was screened (https://osf.io/bdu73/), together with the coding data dictionary. The essentials are given here; the full protocol, sampling frame, and reliability are in Supplement A.

Following the preregistration, we sampled one journal from each of five areas: *Psychological Science* (general), *Journal of Experimental Psychology: General* (experimental), *Journal of Personality and Social Psychology* (social), *Developmental Science* (developmental), and *Psychological Medicine* (clinical). The five were selected on preregistered discretionary criteria: a first-quartile 2023 Impact Factor, a broad rather than narrowly specialized scope, not primarily publishing reviews, and strong reputation in their field, as jointly judged by the coauthors. From each journal's full set of 2024 articles in Scopus, records were screened in a fixed random order until 40 eligible interaction-testing articles were reached, for a preregistered total of 200. Eligible articles were empirical reports of original research; reviews, meta-analyses, editorials, theoretical and qualitative-only articles, replication-only studies, and psychometric validation studies were excluded. Detailed coding was applied only to articles testing at least one interaction, and only to observed-outcome analyses: articles whose relevant tests were purely latent-variable or structural-equation models fell outside the coding scheme (12 such exclusions).

Each interaction-testing article was independently coded by two researchers, with disagreements resolved by discussion or by consulting the other coauthors. Generative AI tools were used only as an optional aid to locate passages within an article; all coding used in the review is human.

For each article, the coders recorded five things: whether at least one tested interaction used a non-identity link; whether the link function was explicitly reported; whether at least one interaction applied an incorrect identity-link analysis (i.e. applied it to a formally constrained observed outcome); whether at least one such interaction was statistically significant, or interpreted analogously in a Bayesian analysis; and which response-variable types were analyzed. By a formally incorrect identity-link analysis we mean a Gaussian linear regression with an identity link applied to an outcome with formal constraints, such as bounds, discreteness, positivity, or a known chance level. This is a conservative screening flag, applied only when the constraint was identifiable from the report; it is not an error rate and does not imply the substantive conclusion is wrong. The full operational rules the coders followed, including borderline cases, are tabulated in Supplement A. Interrater agreement on the double-coded records was high (Cohen's κ between 0.66 and 0.81); per-variable agreement and coder margins are reported in Supplement A. Preregistered variable definitions, operational coding rules, variable-specific agreement, and by-journal results are also reported in Supplement A, Section S2.

Review Findings

First, interaction testing was common. The review identified 331 eligible empirical articles, and 200 of these articles tested at least one interaction, corresponding to 60.4%, 95% CI [55.1, 65.5]. This confirms that interaction testing is a routine part of empirical reporting in the sampled journals.

Second, model-scale reporting was uncommon. Among the 200 interaction-testing articles, 58 used at least one non-identity link function, 29%, 95% CI [23.2, 35.6], and only 46 explicitly reported the link function, 23%, 95% CI [17.7, 29.3]. Thus, in about 77.0% of interaction-testing articles, the link function was not explicitly stated.

Even the 46 explicit cases rest on a generous coding rule. We counted a link-specific model name as an explicit declaration, because such names identify the link: "logistic regression" names the logit link, “probit regression” the probit, and “Poisson regression” the log. Naming only a family or a “generalized linear model” was not enough. Most of the explicit cases arose this way, from the model’s name rather than from a statement of the link in words; had we required the link to be named directly, the count would be lower still.

Third, constrained outcomes were often analyzed on the identity scale. Among the 200 interaction-testing articles, 144 included at least one formally incorrect identity-link analysis of a constrained observed outcome, 72%, 95% CI [65.4, 77.8]. As noted, this is a conservative screening count, flagged only when the constraint was clear from the report. Of these 144 cases, 124 reported at least one statistically significant interaction, 86.1%, 95% CI [79.5, 90.8]; that is, 62.0% of all interaction-testing articles combined a constrained outcome, an identity-link analysis, and a significant interaction claim. This is the same concern raised by work on outcome-model mismatch in interaction testing (Domingue et al., 2024; Liddell & Kruschke, 2018). Table 1 summarizes these results.

The pattern was present across the sampled journals. In Supplement A, the proportion of formally incorrect identity-link analyses of constrained observed outcomes among interaction-testing articles ranged from 40.0% in Psychological Medicine to 95.0% in Psychological Science. In this sample, Psychological Medicine had the lowest screening proportion and the highest non-identity-link proportion (40.0%). This journal, which uses many outcomes for which non-identity links are standard, contributed the fewest formally incorrect identity-link cases.

The outcomes used in interaction analyses were heterogeneous. The most common broad, non-exclusive categories were binary, accuracy, or proportion outcomes (84 articles), sum scores or composites (66 articles), ordinal, Likert, or rating outcomes (36 articles), response times or durations (30 articles), neural or physiological measures (23 articles), difference or distance scores (17 articles), counts or error counts (9 articles), and correlations or associations (8 articles). This heterogeneity matters because the identity link has different implications across outcome types. Together, binary, accuracy, or proportion outcomes, sum scores or composites, and ordinal, Likert,

Table 1

Descriptive summary of the preregistered review. Formally incorrect identity-link analyses of constrained observed outcomes are cases where the reported outcome had formal constraints that make identity-link additivity formally incorrect as a generative model unless explicitly justified as an approximation or as the target observed-score metric.

Quantity	n	Denom.	%	95% CI
Eligible empirical articles	331	331	100.0%	[98.9%, 100.0%]
Eligible empirical articles testing at least one interaction	200	331	60.4%	[55.1%, 65.5%]
Interaction-testing articles using at least one non-identity link	58	200	29.0%	[23.2%, 35.6%]
Interaction-testing articles explicitly reporting the link function	46	200	23.0%	[17.7%, 29.3%]
Interaction-testing articles with formally incorrect identity-link analyses of constrained observed outcomes	144	200	72.0%	[65.4%, 77.8%]
Formally incorrect identity-link cases with at least one significant interaction	124	144	86.1%	[79.5%, 90.8%]
Eligible empirical articles not testing interactions	131	331	39.6%	[34.5%, 44.9%]

or rating outcomes appeared in 152 of the 200 articles (76.0%). The worked examples therefore focus on binomial and aggregated responses, including sums of binary or ordinal items. We thus develop simulation scenarios to show the consequences of link misspecification: many observed psychological variables are built from binary or ordered response events (be them proportions, ratings, averages, totals, or composites). For some approximately continuous outcomes, additivity on the observed scale may be a defensible target. For bounded, discrete, ordinal, chance-constrained, or heavily transformed outcomes, however, additivity on the observed scale is an assumption that should usually be stated and examined. This is especially clear for ordinal and Likert-type outcomes, where treating ordered categories as metric can affect estimates, error rates, and interaction conclusions (Bürkner & Vuorre, 2019; Liddell & Kruschke, 2018).

Table 2

Broad response-variable categories in interaction-testing articles. Categories are non-exclusive, so one article can contribute to more than one row.

regression") rather than the link function per se. Third, constrained outcomes are routinely analyzed on the identity scale: 72% of interaction-testing articles applied an identity-link analysis to an outcome with formal constraints, and most of these reported a significant interaction. In leading journals, then, interaction tests are common while the scale on which no interaction is assumed is rarely stated.

The review documents how analyses were reported. A descriptive review cannot establish how often published interactions are scale artifacts, because that judgment requires knowing the scale on which the data-generating process is additive. The remainder of the paper therefore turns to settings where the generating process is known by construction.

The Link Function Problem in Three Common Cases

To demonstrate the consequences of link misspecification we conduct three simulation studies:

1. *Forced-Choice Accuracy*: This case holds the binomial outcome family fixed and evaluates whether the link function must explicitly encode a task-implied chance floor (e.g., a 0.50 guessing threshold).
2. *Within-Family Link Choice (Logit vs. Probit)*: This case holds both the binary family and the 0-to-1 bounds fixed, testing whether choosing between these two standard link functions alters the results.
3. *Sum Scores*: This case adds the target metric as a third choice alongside family and link. A total built from item responses is itself a measurement mapping, so an analysis of the manifest score tests additivity in raw-score units, even though the substantive claim is generally made about the underlying construct.

Each case develops the conceptual problem and then reports a simulation that quantifies it. The three simulations ask whether a purely additive data-generating process on one scale is detected as a statistically significant interaction when modeled on another scale. A final simulation, reported in the next section, examines whether standard diagnostics reliably identify the misspecified link. The simulation scenarios were chosen to represent plausible psychological data patterns. Each reproduces a common interaction-testing situation: two main effects that are well established and pretty large, for example a group difference combined with a condition effect or a developmental trend, and the question of whether one effect moderates the other. The two main effects were always tuned strong enough to push expected values close to a floor, ceiling, or chance level while keeping them clearly inside the observed range. This is the region where link curvature acts most strongly.

Each simulation applies the same design. Data are generated with no product term on the generating scale, and models assuming additivity on that scale and on plausible alternative scales are fitted to the same data. Every fitted model yields a product-term rejection rate. Rejection under a matched generating-scale model is the nominal rejection rate, whereas rejection under a mismatched-scale model is the pseudo-interaction detection rate.

Forced-Choice Accuracy and the Non-Zero Chance Floor

If a participant answers Y_i of k_i trials correctly, a binomial sampling model, $Y_i \sim \text{Binomial}(k_i, p_i)$, is the natural starting point. The family fixes the sampling process, but not the lower bound that is meaningful for the task. In an m-alternative forced-choice task, guessing yields an expected accuracy of $c = 1/m$, which is .50 in a two-alternative or true/false task, so a claim about performance above guessing places the lower asymptote of the performance curve at c, not at 0.

Three specifications treat this floor differently. A Gaussian identity model on the raw proportions ignores the trial process, assumes additivity in percentage-correct units, and can predict outside the 0-to-1 range. A standard binomial logit or probit model, $p_i = F(\eta_i)$ with F the logistic or standard-normal CDF, respects the sampling process and the 0-to-1 bounds and places the lower asymptote at 0. A chance-corrected binomial link keeps the binomial family and maps the linear predictor onto the interval from chance to 1,

$$p_i = c + (1 - c)F(\eta_i),$$

which for $c = .50$ becomes $p_i = .50 + .50\,F(\eta_i)$; equivalently, the link acts on the above-chance quantity $(p_i - c)/(1 - c)$. A interaction term in the standard model tests additivity over the full 0-to-1 range, and a interaction term in the chance-corrected model tests additivity on a performance-above-chance scale that already encodes the task floor.

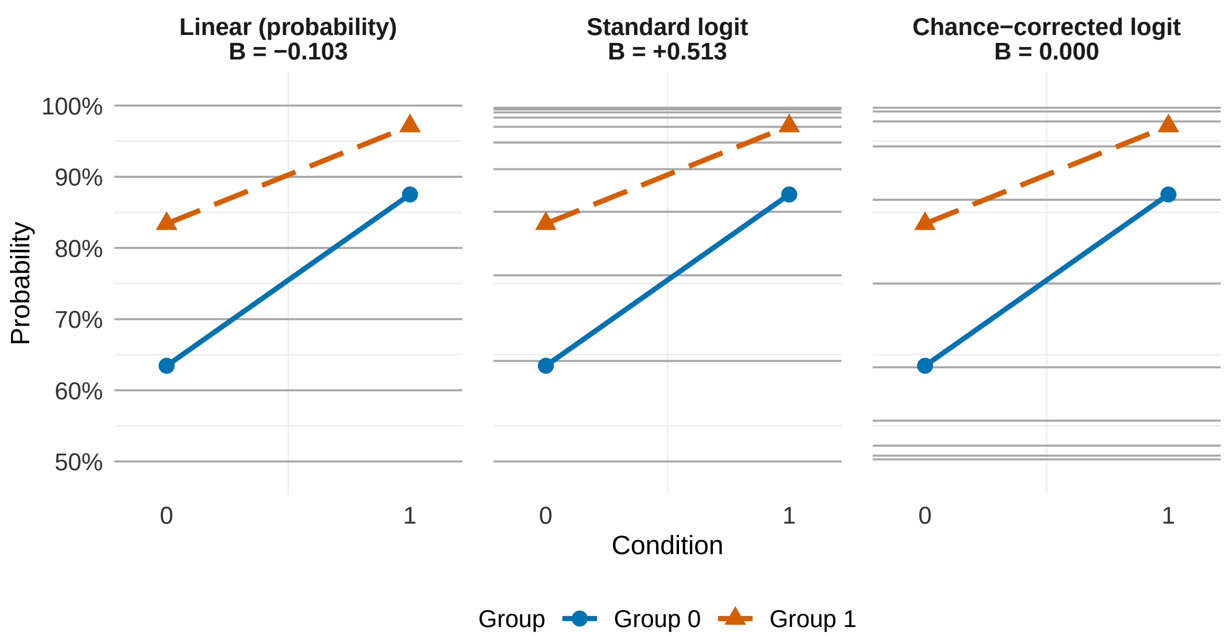
The mismatch is most severe when one group or condition lies near chance. Suppose age has the same effect in two groups on the performance-above-chance scale, with no true age-by-group interaction term there. Near .50 the chance-corrected curve has little response-scale room, so the lower-performing group shows stronger compression on the observed accuracy scale. A standard logit or probit link puts that compression near 0 in place of near .50. The age-by-group difference between differences is exactly zero on the chance-corrected scale and systematically nonzero on the standard logit scale, so the standard model detects a genuine property of its own scale (Figure 2).

The chance-corrected link is most defensible when the chance level is known from the task design, guessing is a plausible lower-bound process, and below-chance performance is not substantively meaningful for the target claim. In the simplest case, the chance level can be fixed by design, for example $c = 0.50$ in a 2-AFC task. In other cases, analysts may estimate the lower asymptote or report sensitivity analyses over plausible values of c. Lapses, response biases, item heterogeneity, changes across trials, or systematic below-chance performance may instead require richer psychometric solutions such as item-response models (e.g., Embretson & Reise, 2000; Wichmann & Hill, 2001).

The chance-corrected GLM link used here closely relates to psychometric-function models (Knoblauch & Maloney, 2012; Wichmann & Hill, 2001) that estimate the chance process at the level where it actually operates. For example, the 3-parameter logistic Item Response Theory model includes an item-specific guessing parameter rather than a single chance level fixed at the design level (Embretson & Reise, 2000). The GLM link should thus be seen as a design-level approximation, not a substitute for item-level measurement models when the data and scientific question call for that additional structure.

Figure 2

A minimal 2×2 *design crosses a condition (x-axis) with a group (color), shown on three scales. The four cell probabilities are identical in every panel; only the scale on which the interaction is evaluated changes. The data were generated with no condition-by-group interaction on the chance-corrected logit scale, which has a* 0.50 *chance floor, so the interaction coefficient is exactly zero on that scale. The same cells imply a positive interaction on the standard logit scale and a negative interaction on the raw probability scale. Light horizontal lines are equally spaced on each panel's own link scale, so equal steps on that scale map to unequal steps on the probability axis.*



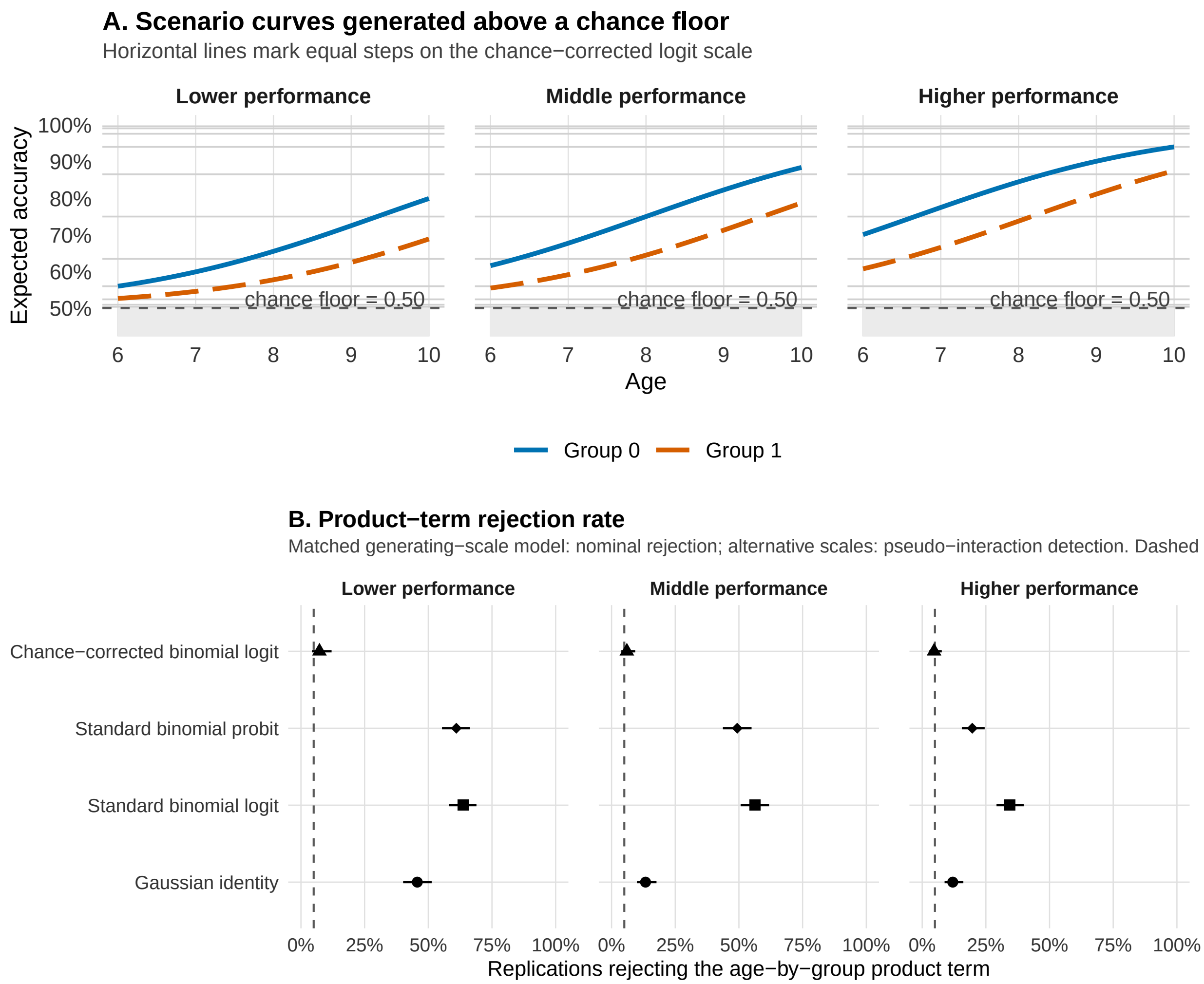
Simulation 1: Design and Results

In each of 300 replications we generated accuracy for $N = 250$ participants with $k = 20$ trials each, from a chance-corrected logit model with a .50 floor, a positive age effect, a group difference, and no age-by-group product term on the generating scale. Three scenarios differ only in overall performance, set by shifting the intercept: in the *lower-performance* scenario the two groups lie close to the .50 floor (expected accuracies roughly .53 to .80 across ages 6 to 10), in the *middle-performance* scenario they occupy the middle of the above-chance range (roughly .55 to

.88), and in the *higher-performance* scenario they approach the ceiling (roughly .61 to .94). Each simulated dataset was analyzed with four aggregate models, one row per participant: a Gaussian linear model on the observed proportions, standard binomial-logit and binomial-probit GLMs on the correct-response counts, and the chance-corrected binomial-logit GLM that matches the generating scale. We recorded how often each product-term test rejected at $\alpha = .05$. The matching model supplies the nominal rejection rate, while the three alternative scales supply pseudo-interaction detection rates. Figure 3 shows the simulation results; the per-scenario rates are tabulated in Supplement A.

Figure 3

Simulation 1: forced-choice accuracy with a chance floor. Data are generated with no age-by-group product term on the chance-corrected logit scale. Panel A shows the resulting accuracy curves across three performance levels; the grey region lies below the .50 chance floor. Panel B shows product-term rejection rates by fitted model and scenario. The matched chance-corrected model remains near nominal alpha, whereas the alternative scales produce pseudo-interactions. The dashed line marks alpha = .05.



The chance-corrected model stays near the nominal .05 in every scenario, with nominal rejection rates between 5% and 7%, because it matches the generating scale. Pseudo-interaction detection is highest where performance lies near the floor: in the lower-performance scenario the standard logit rejects 64% of the time, the standard probit 61%, and the Gaussian identity 46%, against the nominal 5%. Detection falls as performance moves away from the floor, reaching 34% for the logit and 20% for the probit in the higher-performance scenario. When the true floor is .50 and the fitted link places it at 0, the product term compensates for the wrong no-interaction baseline. The same qualitative sensitivity to the chance floor also appears at chance levels 0.25 and 0.33 (Supplement B).

The chance-corrected model can be numerically demanding. Near the floor the likelihood carries little information about changes on the above-chance scale, and some fits fail to converge; in the lower-performance scenario, only 179 of the 300 replications did. This instability is itself informative, signaling that the design gives limited evidence about above-guessing differences in that region. Weakly informative priors can help stabilize estimation when binomial data provide limited information (Gelman et al., 2008), although the added assumptions should be stated and checked.

Within-Family Link Choice

The forced-choice case involved a link that was plainly wrong for the task, because it placed the lower asymptote at 0 when the design fixed it at chance. But the link-function problem can also arise in less obvious cases. Logit and probit are both standard for binary data, respect the same 0-to-1 bounds, and usually produce very similar fitted probabilities. Their choice is therefore often guided by convention, ease of interpretation, or implementation, with little expected impact on substantive conclusions (Agresti, 2002; Yu & Huang, 2019). This makes them a special case for illustrating the problem, as even two virtually interchangeable links define slightly different scales of additivity. Equal differences under one link are not exactly equal under the other, especially toward the tails, and this alone can alter the estimated inte

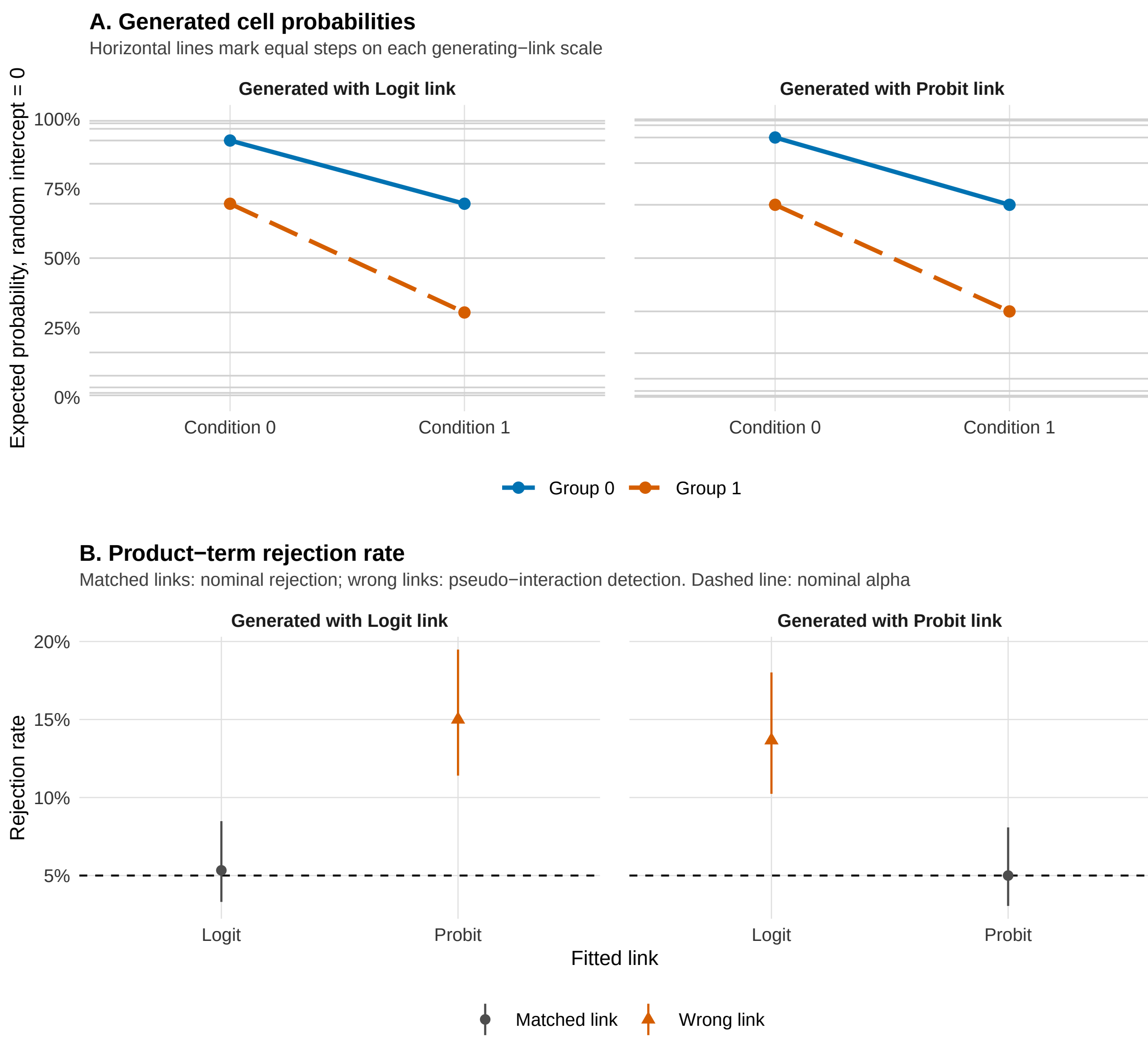
Both links can be represented as latent-threshold models. They differ in the assumed latent error distribution, logistic for the logit link and standard normal for the probit link, and therefore in curvature and tail behavior. The logit parameterization also gives coefficients a direct log-odds interpretation, whereas probit coefficients are expressed on a standard-normal latent scale, making it a choice of election when modeling psychological quantities such as ability, signal strength, response tendency, etc. (Dunn & Smyth, 2018).

Simulation 2: Design and Results

In each of 300 replications we generated repeated binary responses for $N = 700$ participants in a two-group, two-condition design, with 15 trials per participant-condition cell and a subject random intercept (latent intraclass correlation 0.30). The quite large sample was chosen to make the relatively small systematic discrepancy between logit and probit links detectable, rather than to represent a typical psychological study. The condition-by-group interaction term was zero on the generating link scale. We crossed the generating link with the fitted link, using logit and probit in both roles, and fitted every model as a random-intercept GLMM, recording how often each returned a significant condition-by-group interaction term at $\alpha = .05$. Figure 4 shows the results; the full crossed-link rates are tabulated in Supplement A.

Figure 4

Simulation 2: logit versus probit in repeated-trial random-intercept GLMMs. Data are generated with no condition-by-group product term on the generating link scale. Panel A shows the corresponding cell probabilities under logit and probit links. Panel B shows product-term rejection rates for each generating–fitted link combination. Matched links remain near nominal alpha, whereas wrong links produce pseudo-interactions. Error bars are Wilson 95% confidence intervals; the dashed line marks alpha = .05.



When the fitted link matches the generating link, the nominal rejection rate stays near .05:

5% for logit fitted to logit-generated data and 5% for probit fitted to probit-generated data. Crossing the links raises the pseudo-interaction detection rate to 14% when a logit model is fitted to probit-generated data and 15% when a probit model is fitted to logit-generated data, roughly three times the nominal rate. The wrong-link detection rates are smaller than in the forced-choice case, and they appear among links that share the same family, the same 0-to-1 bounds, and nearly overlapping fitted curves in the middle of the probability range, the setting many applied analysts treat as interchangeable. Under the wrong link the same cell probabilities imply a small but deterministic nonzero product term on the fitted scale. Supplement B shows that its detectability varies non-monotonically with latent location, becoming small in some regions and much larger in others. A result that is stable across both links is easier to defend; one that appears only under a single link should be reported as link-conditional, or translated into a response-scale estimand and checked across plausible links. The deterministic product term induced by evaluating each generating model on the alternative link scale is reported in Supplement A, Section S5.

Sum Scores

Many psychological outcomes are computed as total sum scores across questionnaire items, task items, symptoms, errors, or ratings. In the two cases above the outcome was a directly observed response, and only the link was in question. A sum score is different, because the score itself is already the result of an analytic choice. What is observed are the individual item responses, which are binary or ordinal; the total is a summary of them, produced by a scoring rule that gives every item the same weight and every point of the scale the same width. In psychological research the construct of interest is often the underlying, generally continuous dimension, so the total can be seen as a remapped version of that continuum. The link question then arrives together with a prior question about which metric the claim is about.

That remapping is not linear. The bounds are crucial here, because the construct may have no floor or ceiling while the observe total score has both. The mapping is best read as an exchange rate between the two scales, that is, how much of the latent dimension is needed to gain one raw-score point. That rate is not constant. Near the middle of the scale a small latent shift is enough to move the total by a point, while near either bound a wide stretch of the latent dimension is compressed into the last available point, because few items remain that the respondent can still change. To illustrate, with nine items scored 0-3, moving from 13 to 14 points requires less change on the latent dimension than moving from 26 to 27. A constant latent effect therefore produces smaller raw-score differences as the total approaches an end, and the same reasoning applies wherever the instrument is unevenly informative along the latent dimension, so compression is not confined to the endpoints.

Bounded, discrete, and skewed distributions are common in psychological data (Micceri, 1989), and the consequences of analysing ordinal responses as if they were metric have been documented in detail (Liddell & Kruschke, 2018). Fitting a Gaussian identity model to a total assumes instead that a predictor moves the score by the same number of points at every level of the other predictor. A group/condition sitting near a bound moves by fewer points even when the data-generating process on the latent dimension is the same in both groups/conditions. Behaviour-genetic research has met the same problem when testing whether genetic effects vary across environments, since those interactions are usually tested on summed questionnaire scores: analysed at the item level, the same data show that the total can conceal real interactions or present spurious ones through properties of the observed score (Molenaar & Dolan, 2014). This is the sum-score form of outcome-model misspecification and target-metric mismatch (Domingue et al., 2024).

The problem we raise sits next to a recent, wider debate about whether sum scores should be replaced by factor scores or latent-variable models. Some argue that sum scores are transparent, reliable enough for many purposes, comparable across studies, and tied to how instruments are scored in applied settings (Widaman & Revelle, 2023); the counterargument is that a total encodes strong and usually untested assumptions about how items represent the construct (McNeish, 2023; McNeish & Wolf, 2020). While we do not enter the debate, we add a narrower point that was left aside: a model of the manifest total defines the absence of an interaction in raw-score points, so an interaction found there is an interaction on the score, and it becomes a claim about the construct only under the additional assumption that the two are linearly related.

When item-level data are available and the claim concerns latent constructs, an explicit measurement model such as SEM, item-level ordinal regression, or IRT-inspired specifications could be preferred, since it addresses the mapping directly, though at the price of moving the assumptions into the measurement model (Bürkner & Vuorre, 2019). Often, however, researchers deliberately work with the total score for the practical reasons listed above. An option is to set a family that respects the bounds and the discreteness, for example modelling the total as a bounded count with a beta-binomial likelihood (Wilcox, 1981). However, we suggest a deliberately unusual option that moves only the link: keep the observed total score under a Gaussian likelihood, but rescale the score to the proportion of the maximum and use a probit link function, as in Simulation 3 below. This is an *ad hoc* device, not a standard practice, but it illustrates whether the interaction conclusion changes if additivity is placed on a nonlinear scale that compresses the expected values toward the score bounds. It remains a model of the total, it will not recover the construct, and the Gaussian likelihood is still misspecified in its variance, since a bounded score has a variance that depends on its mean: what the probit link corrects is only the scale on which additivity is defined, not the distributional form. Read against the sum-score debate, it is a precaution for researchers who have decided to analyse totals anyway, but acknowledge that the interaction still needs a scale on which additivity is plausible.

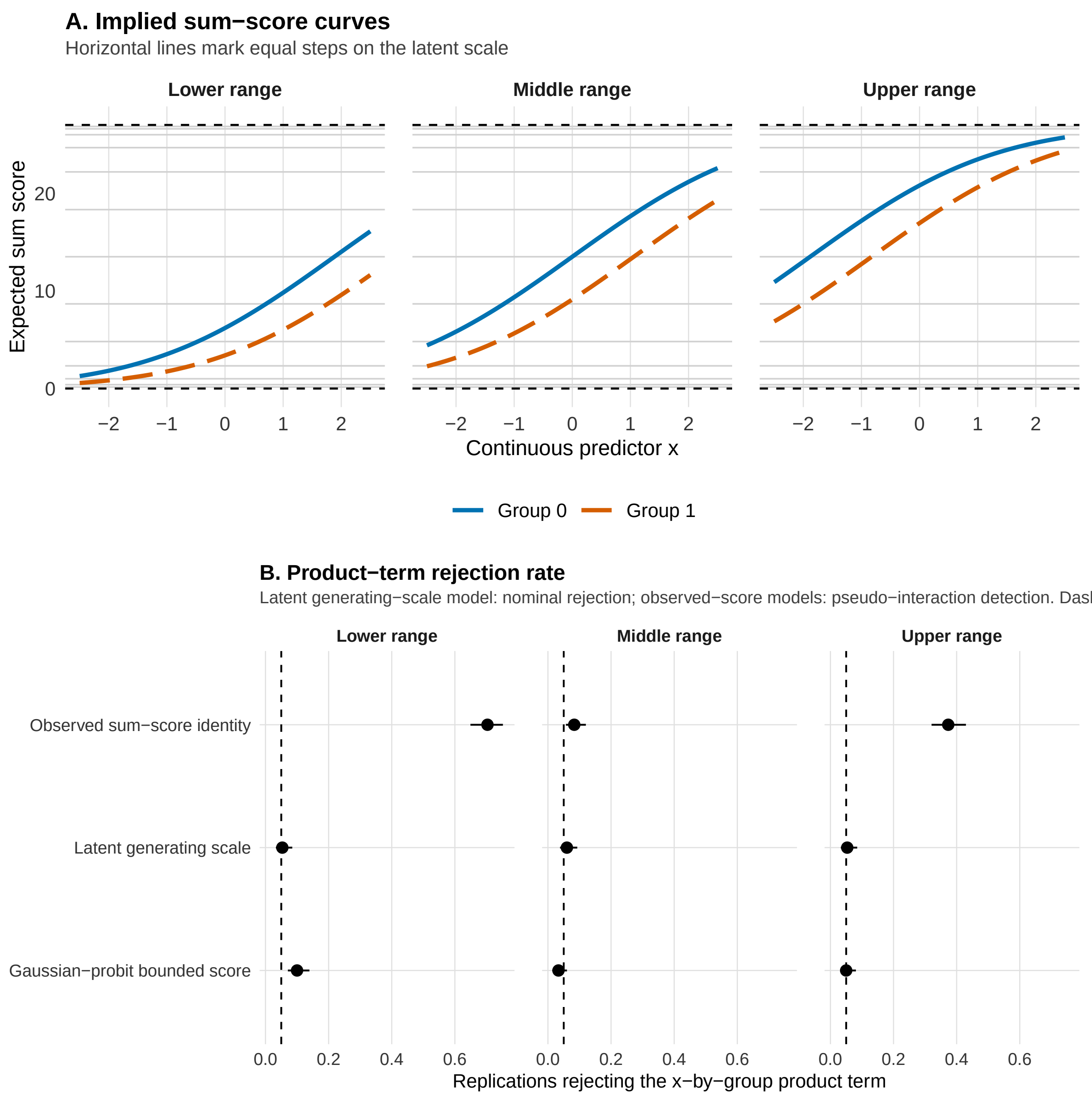
Simulation 3: Design and Results

The bounded-score probit model introduced above is tested here against the two natural alternatives: the manifest total and the latent scale it approximates. In each of 300 replications we generated data for $N = 600$ participants on a latent continuous scale, with a continuous predictor, a group effect, and no true x-by-group interaction term. Each latent value was converted into 9 ordinal items scored 0 to 3, which were summed to a total on a 0-to-27 scale. Three scenarios shift the item thresholds so that expected totals fall in the middle of the scale, near the floor, or near the ceiling. Each participant's total score was analyzed on three scales, chosen to isolate the relevant comparison: the observed sum-score identity model, a Gaussian linear model of the manifest total; the Gaussian-probit bounded-score model, a sensitivity model whose inverse link constrains the rescaled conditional mean within the bounds of minimum-maximum range; and the latent generating scale itself, an idealized benchmark available only because the true theta is known. We recorded how often each returned a significant x-by-group interaction term at $\alpha = .05$. Figure 5 shows the results; the per-scenario rates are tabulated in Supplement A, Table 8.

The bounded-score model fits a continuity-corrected score proportion with a Gaussian likelihood and a probit link, then rescales predictions back to the sum-score metric. A probit rather than a logit link is used because its standard-normal linear predictor keeps the ad hoc correspondence to a latent scale simple and readable as if they were standardized units (but note that, as we showed in a previous section, this is *not* a neutral choice).

Figure 5

Simulation 3: sum scores as bounded and discrete outcomes. Data are generated with no x-by-group product term on the latent scale. Panel A shows how the same additive latent structure maps onto sum scores in the middle of the scale, near the floor, and near the ceiling. Panel B shows product-term rejection rates for the latent benchmark and two observed-score models. The latent model remains near nominal alpha, whereas the observed-score models produce pseudo-interactions. The dashed line marks alpha = .05. The latent benchmark is available only in simulation.



The latent generating scale, the idealized benchmark, has nominal rejection rates near .05 in every scenario (between 5% and 6%), confirming calibration on the scale that generated the data. The observed sum-score identity model has a similar pseudo-interaction detection rate only in the middle of the scale, where the total is approximately linear in the latent variable (8%). As expected totals approach a bound, its detection rate rises sharply, reaching 37% near the ceiling and 70% near the floor, because uneven compression of the latent scale makes a constant latent group effect appear to change across levels of the predictor. The Gaussian-probit bounded-score model has pseudo-interaction detection rates close to alpha throughout (between 3% and 10%). This sensitivity model respects the score bounds but is not the true latent measurement model.

The pattern is conditional on the region of the score range. Over the ranges examined in Supplement B, scale location changed identity-model rejection much more than sample size, item count, or latent dispersion. Far from the bounds, raw-score additivity can approximate the latent-scale result; near a bound, compression can create an interaction even when the construct-level effect is constant. The bounded-score probit model is a useful sensitivity check, and it remains a model of the manifest total. When the claim concerns an interaction between latent constructs, the more direct solution is an explicit measurement model, such as SEM, item-level ordinal, or IRT-inspired models; these still require assumptions about thresholds, links, loadings, and latent-response metrics, so they make the measurement part of the interaction claim explicit without removing the scale problem.

Can Standard Diagnostics Catch a Wrong Link?

The simulations above assume the analyst does not question the fitted model. In practice, a researcher who finds an interaction may run model checks before reporting it. We examine three: a direct link test, simulation-based residual diagnostics, and information criteria. This set represents distinct strategies, testing the assumed link directly, screening for misfit of any kind, and ranking a set of candidate models. Such strategies are checks that applied psychologists might routinely apply. A further strategy, estimating the shape of the link from the data through a parametric family of links, is more direct and rarely used in psychology (Aranda-Ordaz, 1981;

Czado & Santner, 1992; Stukel, 1988). None of the strategies directly identifies the right scale on which the no-interaction claim is defined. At best, they point to the link that fits the data best among a set of alternatives.

A link test of the kind proposed by Pregibon targets link adequacy directly, in the added-variable form used here (Pregibon, 1980). The model is fitted, the fitted linear predictor $\hat{\eta}$ is extracted, and the model is refitted with $\hat{\eta}^2$ added as an extra predictor. If the link is adequate the squared term carries no additional information and should be non-significant; a significant term signals curvature that the current link leaves unexplained. The signal is not specific to the link: an omitted nonlinear term, an omitted covariate, or a missing interaction produce it as well. A non-significant result means only that this diagnostic did not detect a departure in this sample.

Residual diagnostics cover more ground. In a GLM, raw, Pearson, and deviance residuals have no common reference distribution across families, which makes residual plots difficult to read. Randomized quantile residuals tackle this problem via simulation (Dunn & Smyth, 1996): many datasets are generated from the fitted model, and each observed value is located within the distribution of its own simulated values, so a value at the median of that distribution receives .50 and a value in the far tail receives a value near 0 or 1. Under an adequate model these positions are uniform between 0 and 1 for every family, so misfit appears as departure from uniformity, as over- or underdispersion, or as residual structure over fitted values and predictors. We use the implementation in the DHARMa package of R (Hartig, 2024). The wider coverage comes at the cost of specificity, however: link misspecification, wrong family, missing nonlinearity, missing interaction, and target-metric mismatch all appear as the same kind of departure from uniformity.

Information criteria such as AIC, BIC, WAIC, and LOO rank model fit among a supplied set of alternatives. For a link question, the relevant comparison is the following: for the same response data, the same family/sampling model, the same structural formula, what alternative link function demonstrate the best fit? Thus, it is better understood as a targeted sensitivity analysis rather than a model selection. Two limits apply: first, information criteria only compare fit within the candidate set, so this does not identify the scale on which the scientific claim is made; second, the fitted model already contains the product term, and that term can absorb part of the curvature created by the wrong link, so the misspecified model can fit the observed data almost as well as the target model even while it places additivity on the wrong scale. In the comparisons reported here the two candidates always have the same number of parameters, so the ranking rests entirely on the log-likelihood.

Diagnostic Simulation

We simulate five scenarios covering different cases, each being tied to a previous example in the paper. The first is tied to the simple count example in Figure 1. Data are generated from a Poisson-log model in which expected errors decline with age, with no age-by-group interaction term on the log-count scale. The diagnostic scenario uses the same design as the figure, with coefficients recorded in the generated scenario table: intercept 2.20, age effect -0.55 per year after age 6, and group effect 0.45. The fitted model keeps the Poisson family and the interaction formula but uses an identity link. This makes the comparison a same-family link comparison: the family and the formula are held fixed, and only the link changes. In real research scenarios, however, family and link often covary.

The second scenario is the lower-performance forced-choice scenario from Simulation 1. Data are generated with a .50 chance floor, 20 trials per participant, intercept -0.80, age effect 0.60, group effect -0.90, and no age-by-group interaction term on the chance-corrected logit scale. The fitted model is a standard binomial-logit interaction model, which respects the binomial trial structure while placing the lower asymptote at 0, below the task-implied chance level. The chance-corrected fit was initialized from the corresponding standard-logit model and optimized with BFGS, with Nelder–Mead used as a fallback when needed.

The third scenario mirrors the probit-generated, logit-fitted case in Simulation 2. Data are repeated binary trials with a subject random intercept, 15 trials per condition cell, and latent ICC .30. The fixed effects are the probit reference values used in Simulation 2: intercept 1.50, group effect -1.00, condition effect -1.00, and no group-by-condition interaction term. The fitted model is the corresponding random-intercept logit GLMM with the same interaction formula.

The fourth scenario is a continuous-predictor counterpart to the previous binary repeated-trials case. It keeps the probit data-generating link, the logit fitted link, the group effect, the focal-predictor effect, the subject random intercept, and the latent ICC, and replaces the two-level condition with a continuous predictor sampled from 0 to 1. This scenario asks whether the same logit-versus-probit issue is visible when the focal predictor has enough unique values for continuous-predictor residual checks to be better identified.

The fifth scenario uses the Gamma response-time example from the introduction. Data are generated from a Gamma model with a log link, shape 8, baseline mean 400 when x = 0 and group = 0, x effect 0.25 on the log-mean scale, group effect 0.25, and no x-by-group interaction term. The fitted model keeps the Gamma family and the interaction formula and uses the inverse link.

In each replication, diagnostics were applied to the misspecified interaction model, because this is the model an analyst would usually inspect after finding the interaction. This design targets a realistic post-estimation question: does the same fitted model that produced the product-term finding also show enough diagnostic evidence to warn the analyst about the link problem? The choice is demanding because the interaction term can absorb part of the curvature created by the wrong link. A diagnostic workflow applied to an additive model before adding the interaction term would answer a different question. The present simulation asks whether ordinary checks protect the interaction conclusion once the misspecified interaction model has been fitted.

The residual workflow used four DHARMa checks: residual uniformity, dispersion, residual quantile patterns over fitted values, and residual structure over the focal predictor, using a quantile check for continuous predictors and a design-cell comparison for the categorical 2×2 case (Dunn & Smyth, 1996; Hartig, 2024). We also used a Pregibon-style added-term link check as a more link-oriented diagnostic (Pregibon, 1980). As a calibration check, we also applied the same DHARMa and Pregibon-style procedures to the corresponding correctly specified interaction model in each replication. These rates estimate how often each diagnostic signaled a problem when the fitted link was correct.

Finally, we used AIC only for relative comparisons among prespecified candidate mean models fitted to the same response data with the same interaction formula. In the count and Gamma scenarios, the response family is held fixed and only the link changes. In the binary scenarios, the binomial sampling model is held fixed. The chance-floor scenario compares two binomial mean mappings: a standard logit model and a chance-corrected logit model with a design-implied lower asymptote. Thus, AIC varies only the candidate mean-link mapping, and the interaction term is present in every candidate model. We report the proportion of replications in which the lower AIC favored the target model, and also the distribution of the AIC difference, defined as the AIC of the misspecified model minus the AIC of the target model, so that positive values favor the target link. The proportion by itself is not informative enough, because a link that wins most replications by less than one AIC unit and a link that wins by fifty describe different situations. Because no inconclusive ΔAIC category is used, the complement favors the misspecified model when both AIC values are available. These values describe relative fit within a limited candidate set.

Table 3 documents diagnostic misalignment: the misspecification that generates a pseudo-interaction does not produce a matching diagnostic signal, and protection varied with the candidate models supplied. The chance-floor scenario is the clearest case. The pseudo-interaction detection rate was 59%, while all four DHARMa diagnostic detection rates stayed near nominal levels. Among replications in which the wrong link produced a significant pseudo-interaction, all applicable DHARMa checks remained non-significant in 93.8% (166/177). The check that responded was the explicit link comparison, and its margin was thin: AIC favored the chance-corrected model in 78% of replications, by a median of 2.2 AIC units, with the two links within two units in 39.7% (119/300) of the finite comparisons. The candidates thus separated the interaction conclusion much more sharply than they separated fit, and a quiet residual check is not evidence that the link is adequate for the interaction claim. For the two cases extended in Supplement B, pseudo-interaction detection, Pregibon-style detection, and AIC preference also change at different rates across sample size and scale location.

Calibration rates varied across diagnostics or scenarios, so flagging rates under

Table 3

Diagnostic simulation across five link-misspecification scenarios. Pseudo-interaction reports how often the misspecified model detected an interaction that was absent on the generating link scale. DHARMa gives the range across the four prespecified residual checks. Pregibon reports the added-term link check. AIC favors target reports the proportion of replications in which the lower AIC favored the target interaction model over the misspecified interaction model, with the structural formula held fixed. The complementary proportion favors the misspecified model when both AIC values are available. CI = Wilson 95% confidence interval.

Case	Target model	Misspecified model	Pseudo-interaction rate	DHARMa	Pregibon	AIC favors target
Count errors	Poisson log	Poisson identity	0.957 [0.927, 0.975]	0.000-0.883 (4)	1.000	0.986
Chance-floor accuracy	chance-corrected binomial logit	standard binomial logit	0.590 [0.534, 0.644]	0.000-0.050 (4)	0.290	0.783
Binary repeated trials	binomial probit GLMM	binomial logit GLMM	0.113 [0.082, 0.154]	0.000-0.007 (4)	1.000	0.937
Binary continuous predictor	binomial probit GLMM	binomial logit GLMM	0.070 [0.046, 0.105]	0.000-0.023 (4)	1.000	0.940
Gamma mean response time	Gamma log	Gamma inverse	0.307 [0.257, 0.361]	0.000-0.053 (4)	0.353	0.780

misspecification should be read against the baseline rates reported in Supplement A.

Practical Starting Points for Link-Aware Interaction Testing

The applied output of the framework is a small set of outcome-specific starting points. These starting points are prompts for the choices that should be visible in a scientific paper: what link defines the scale of additivity, what metric the claim is about, and what sensitivity check is feasible for the design (Table 4). Sometimes the observed metric is itself the target: a clinician, teacher, or researcher acts on a two-point difference on the reported score, and raw-score additivity is then the quantity of interest. What matters is that this is stated, rather than inherited from a default model. The preferred link should be justified from theory, task structure, or measurement whenever possible (Baranger et al., 2023; Castillo et al., 2026). When these considerations do not uniquely determine the link, the interaction should be probed through a planned link-sensitivity analysis. Researchers should fit alternative defensible specifications, compute the same target response-scale contrast across them, and state whether the interaction conclusion is stable or link-conditional. When the claim is about the observed response, that contrast (rather than the product-term coefficient) is the quantity that should be reported and compared across links (McCabe et al., 2022; Mize, 2019; Rohrer & Arel-Bundock, 2026). This shows which part of the conclusion depends on modeling conventions and which part does not. Stable conclusions across plausible specifications are easier to defend. Unstable conclusions should be reported as conditional on the modeling choice, as choices that affect the claim should be visible, justified, and interpretable (Mustillo et al., 2018).

Discussion

The same data can support different interaction conclusions under different plausible model specifications. This is the familiar scale dependence of interactions (Cox, 1984; Loftus, 1978; Wagenmakers et al., 2012). In GLMs, this involves the choice of outcome family, link function, and target metric. Our review suggests that these choices are often left implicit, while the simulations show that they can matter even when the fitted model appears broadly reasonable.

More specifically, our review showed that interaction tests are common, link functions are often implicit, and many interaction-testing articles use identity-link analyses on outcomes for which observed-scale additivity is not self-evident. The simulation studies showed the risk of link misspecification, even within an appropriate family, when two main effects are present. Forced-choice accuracy offers a very clear case: the binomial family is correct, but the standard canonical link function misplaces the lower asymptote. The logit-probit comparison then shows that the problem persists even when the response bounds are correctly specified, between links

Table 4

Practical starting points for link-aware interaction testing across common psychological outcomes.

Outcome type	Plausible starting point	Minimal sensitivity check
Forced-choice accuracy	Binomial logit/probit; chance-corrected link when the task floor is part of the estimand.	Compare standard logit/probit and chance-corrected links; report convergence and response-scale contrasts.
Binary / proportion outcome	Binomial logit/probit; consider cloglog for asymmetric processes.	Compare logit/probit; inspect calibration and contrasts near bounds or rare-event regions.
Likert / ordinal item	Ordinal logit/probit; metric model only with a clear observed-scale target.	Compare ordinal and metric summaries; report category probabilities or response-scale contrasts.
Sum score / composite	Gaussian identity when many levels stay far from bounds; otherwise consider item-level, ordinal, bounded-count, or bounded-score models.	Inspect bounds, skew, and score concentration; compare with item-level or bounded-score sensitivity models when feasible.
Counts / error counts	Poisson or negative binomial, usually with log link; identity only with an observed-count target.	Check overdispersion and zero inflation; compare count contrasts across plausible count models.
Response time / skewed positive outcome	Lognormal, Gamma log/inverse, Inverse Gaussian, or transformed Gaussian, matched to absolute, proportional, reciprocal, or transformed change.	Compare raw-, log-, and response-scale summaries; inspect tail fit and predicted values.

that look similar over most of the range. Sum scores extend it to the target metric itself, where the manifest score is already a measurement mapping from item responses, and where even a likelihood suited to a bounded count leaves the scale of additivity to be chosen.

These results extend previous work showing that interaction effects are often difficult to estimate precisely even when the model form is correct (Baranger et al., 2023; Castillo et al., 2026). Here, the difficulty is specific to link, which can produce statistically significant interaction terms even when the true interaction is zero on the generating scale. The diagnostic simulation suggests that standard model checks do not reliably reveal when an interaction is driven by link choice. Residual checks, Pregibon-style tests, and information criteria were useful in some settings, but their performance depended on the candidate set. The chance-floor scenario is the clearest case: a wrong link produced a large pseudo-interaction while residual checks showed no sign of misfit. There the check that responded was the explicit comparison between the two candidate links, and it separated them by a margin much smaller than the distance between the two interaction conclusions. This is diagnostic misalignment, not uniform diagnostic failure, and it suggests that diagnostics should be treated as evidence about model fit within a candidate set, not as proof that an interaction claim is robust to link choice (Dunn & Smyth, 1996; Hartig, 2024; Pregibon, 1980).

Four limitations bound the scope of the conclusions. First, the review was descriptive and limited to five reference journals across major areas of psychology. Its numerical estimates should therefore be read as reflecting reporting practice in high-standard outlets during the sampled year. A broader mapping across journals, years, and subfields could later support reporting guidance on interaction tests with constrained outcomes. Second, the review coded directly observable reporting features rather than judging, for each article, whether its interaction was scale-dependent enough to be removable. Third, the simulations are mechanism demonstrations aimed at a specific regime, two established main effects that jointly expose observations to link-function curvature. Supplement B broadens these scenarios through structured sensitivity analyses, but it still does not estimate prevalence or cover all realistic complications. Finally, logit and probit often lead to very similar conclusions, especially away from boundaries, and for prediction, the link is largely a computational convenience, and what matters is forecast accuracy and staying within admissible bounds, such as probabilities between 0 and 1; our claim is that link choice can matter for interaction claims, not that it always does.

Interaction effects are routinely reported as if the choice of modeling scale had no substantive consequences. Link-aware testing corrects this by separating three judgments that are often run together: which outcome family is plausible, which link defines the scale on which additivity holds, and whether the conclusion survives plausible alternatives to that choice. This reframing clarifies what kind of claim researchers can defend. If the result is stable across links, the claim can be about the interaction itself; if it is not, the claim is narrower, but still defensible when stated as an interaction on a particular scale rather than treated as a failure.

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